TECHNICAL REPORT #9

MONITORING EMERGENT LITERACY DEVELOPMENT OF IMMIGRANT PRESCHOOLERS WHO SPEAK SOMALI, SPANISH, OR HMONG

THERESA L. ESTREM
DEPARTMENT OF COMMUNICATION SCIENCES AND DISORDERS
ST. CLOUD STATE UNIVERSITY
ST. CLOUD, MN
AND
SCOTT R. MCCONNELL
CENTER FOR EARLY EDUCATION AND DEVELOPMENT
AND
DEPARTMENT OF EDUCATIONAL PSYCHOLOGY
UNIVERSITY OF MINNESOTA
MINNEAPOLIS, MN

TABLE OF CONTENTS

Acknowledgements 3

Abstract 4

Tools for Monitoring Emergent Literacy Development 6

Methods 8

Participants 8

Measures 9

Procedures 11

Results 12

Analysis Strategy 12

Null Model: Emergent literacy skills 14

Conditional Model 1: Emergent literacy skills and *Home Language* 15

Figure Legend 18

Discussion 22

References 27

Appendix 32

Null Model Equations 32

ACKNOWLEDGEMENTS

The authors thank our colleagues at PICA Head Start who collaborated with CEED in developing goals and dreams of optimal learning among all involved: teachers, parents, children, researchers, and administrators. We offer special thanks to Dr. Kristen Missall for her valuable contributions, and to Dr. Michael Rodriquez for his assistance with data analysis and interpretation. We also acknowledge the contributions of the late Dr. Mary McEvoy, who did tremendous work to facilitate the project reported here.

ABSTRACT

A growing number of families served in early childhood programs are immigrants whose primary language is not English. Children from those families may be especially at risk of being less prepared to learn to read, and would benefit most from close monitoring of skill development that supports effective intervention. Because early literacy skills are a commonly recognized precursor to young children's later school success, close monitoring of progress in early literacy skills also is needed. Yet, we have limited descriptive information about the early literacy skill development over time for these English Language-Learning Head Start children. Using Individual Growth and Development Indicators (IGDIs) to repeatedly assess the early literacy skills (expressive language and phonemic awareness) of over 2300 preschoolers enrolled in Head Start, this study describes the developmental trajectories of English expressive language and phonemic awareness skills of boys and girls in several language groups. Results of analyses with Hierarchical Linear Modeling indicated three IGDI measures are sensitive to growth over time, although indicators of phonemic awareness often do not show growth until 48 months of age. The developmental trajectories of early literacy skills were affected by primary language spoken at home. This research has implications for assessing and intervening with early literacy skills for children most at risk for future school success, and for further refinement of IGDIs to address these activities.

MONITORING EMERGENT LITERACY DEVELOPMENT OF IMMIGRANT PRESCHOOLERS WHO SPEAK SOMALI, SPANISH, OR HMONG

Many children who experience reading failure are from minority groups, including children living in poverty (Adams, 1990; Beaulieu, 2002; Nicholson, 1997; Whitehurst, 2000) and children learning English as a second language (Adams, 1990; Snow, 1999; Tabors, 1997, 1998). The number of immigrants in the United States is rising steadily and our schools serve an increasing number of children whose primary language is not English. In fact, in 2003, 19% of school-age children in the United States spoke a language other than English at home (ChildStats.gov, 2005).

Because many immigrant families live in poverty, Head Start serves many young children whose home language is not English. President Bush's initiative, "Good Start, Grow Smart," of the No Child Left Behind Act (The White House website, 2001) mandated Head Start programs to prepare these children for school success, with a special focus on language and literacy development. In addition, a major emphasis of "Good Start, Grow Smart" encourages a strong Federal-State partnership among agencies who provide early childhood education (including Head Start) to ensure pre-reading skills at school entry through accountability and being able to demonstrate children's progress and development toward school readiness, including emergent literacy skills (The White House website, 2001).

Head Start programs typically have administrators, parents, teachers, and teacher assistants who provide support and implement activities designed to facilitate the development of emergent literacy skills. Speech-language pathologists are also increasingly involved in screening, assessment, and interventions related to emergent literacy (Fey, Catts, & Larrivee, 1995; American Speech-Language-Hearing Association, 2001). With limited resources and heavy caseloads, educators must be efficient and diligent when they are with the children and when they perform their record-keeping tasks. This can be accomplished with a measurement tool that is quick and easy to administer, has proven psychometric properties for validity and reliability, can be administered repeatedly, and is sensitive to growth over relatively short periods of time.

There is, in fact, such a measurement tool, called the *Individual Growth and Development Indicators* (IGDIs) for Emergent Literacy, which has been validated for typically developing children, children in poverty, and children with speech and language delays. We do not know whether these IGDIs are sensitive enough to measure the sometimes delayed and slower progress of preschoolers who are learning English as a second language, and whether the IGDIs are sensitive to differences in skills of children who

speak different languages. To address this issue, we conducted a longitudinal study that examines the sensitivity of the IGDIs to measure emergent literacy skills of a large sample of Somali- Spanish-, Hmong-, and English-speaking preschoolers.

TOOLS FOR MONITORING EMERGENT LITERACY DEVELOPMENT

Well-accepted theoretical models consistently indicate that components of emergent literacy include phonological awareness skills (e.g. rhyming, alliteration, sound blending), oral vocabulary and language, print principles, and emergent writing (Snow, Burns, & Griffin, 1998; Whitehurst & Lonigan, 2002). Typical assessments of emergent literacy skills for children between 3- and 5-years of age (e.g. Assessment of Literacy and Language, Lombardino, Lieberman, & Brown, 2006; The Phonological Awareness Test, Robertson & Slater, 1997; Test of Preschool Early Literacy, Lonigan, Wagner, Torgeson, Rashotte, 2007; Test of Early Reading Ability-Third Edition, Reid, Hresko, & Hammill, 2001) have several limitations. First, as with most developmental assessments of young children, they are conducted during single sessions (Neisworth & Bagnato, 1996) and results provide limited or no information about a child's development *over time*. Most notably, the results of these assessments do not include on-going (e.g. monthly) information about a child's development. This lack of repeatability greatly limits professionals' ability to monitor growth and/or improve the effects of treatment (Deno, 1997). Second, assessment of emergent literacy skills does not necessarily lead to intervention of these skills, as it should do in early childhood (Bricker, Pretti-Frontczak, & McComas, 2002; Good, Simmons, & Smith, 1998). In other words, a decision-making model has not been implemented for young preschool children that links on-going assessment of emergent literacy skills with a plan, whether it is to monitor a child's development, provide intervention, or modify intervention strategies (Early Childhood Research Institute for Measuring Growth and Development (ECRI-MGD), Technical Report 5, 1998).

General Outcome Measurement (GOM) addresses the issue of assessing change in a student's performance in a targeted instructional area, such as language and literacy, by administering a task repeatedly over time (Deno, 1986, 1997; Deno, Mirkin, & Chiang, 1982). By definition, GOMs (1) are *easy* to administer and interpret, (2) provide direct assessment and are *sensitive* to growth or progress for different ages or across time, (3) provide *useful* information about the need to modify a child's intervention plan, and (4) are supported by empirical evidence of *reliability* and *validity* (Fuchs & Deno, 1991).

The IGDIs used in this study follow the logic of General Outcome Measurement. IGDIs were developed at the University of Minnesota to assess and describe developmental progress of children from

birth to 5 years of age (Early Childhood Research Institute, Technical Report 6, 1998). More specifically, IGDI preschool measures for oral language (i.e. Picture Naming) and phonological awareness skills (i.e. Rhyming and Alliteration) have been developed and are increasingly used in many applied and research settings (McConnell, McEvoy, & Priest, 2002). Picture Naming, Rhyming, and Alliteration IGDIs meet the GOM criteria for ease, usefulness, sensitivity, reliability, and validity. IGDIs correlate highly with standardized language and literacy instruments that must be administered by a professional with training or expertise, are much briefer to administer (1-2 minutes per measure), can be administered repeatedly within short time frames, and measure growth over short periods of time (McConnell, Phaneuf, & Murphy, 2002; McConnell, Priest, Davis, & McEvoy, 2002; Missall, 2002; Priest, McConnell, McEvoy, & Shin, 2000).

Results from Picture Naming, Rhyming, and Alliteration have demonstrated the development of emergent literacy skills for typical preschool children, children attending a Head Start program, and children with disabilities (McConnell, Phaneuf, et al., 2002; McConnell, Priest, et al., 2002; Missall, 2002; Missall, McConnell, & Cadigan, 2006; Priest, et al., 2000). Small-scale studies have assessed the IGDIs' sensitivity in measuring emerging literacy skills in English of children whose first language is Spanish (Missal & McConnell, 2006), and in English and Hmong of children whose first language is Hmong (Nitsiou, 2001). Given the growing incidence of ELL children in our schools, the mounting evidence for the role of preschool language and literacy development on preventing later reading and academic failure, and the development of IGDIs to monitor early language and literacy development, it is important to further examine developmental trajectories for children who are English language learners. In particular, this study will use the IGDI Picture Naming, Rhyming, and Alliteration measures to examine their ability to measure growth of emergent literacy skills in English of preschoolers who speak Somali, Hmong, Spanish, or English. Further, we will examine whether the home language contributes over and above to the development of emergent literacy skills. Specifically, the following questions will be addressed:

1) What are the mean scores and rates of growth for the Picture Naming, Rhyming, and Alliteration IGDIs for Somali-, Spanish-, and Hmong-speaking preschoolers?
2) Are mean scores and rates of growth significantly different among English-, Somali-, Spanish-, and Hmong-speaking groups?
3) Are the Picture Naming, Rhyming, and Alliteration IGDIs sensitive enough to measure emergent literacy growth of children who speak English as a second language?

METHODS

PARTICIPANTS

A total of 2,306 Head Start preschool children (1115 girls, 1042 boys, 149 unidentified) enrolled in 95 classrooms participated in this study. The children were not directly recruited. Rather, extant data was used without identifying information. A large Head Start program in a metropolitan area and the Center for Early Education and Development (CEED) of the University of Minnesota collaborated on a project to train teachers to administer preschool early literacy IGDIs. As one of their outcome-based measures, this Head Start program had teachers administer IGDIs several times during the school year. Although the program's goal was to administer preschool early literacy IGDIs to all children in these Head Start centers three times during the school year, some children were absent on test administration days and/or were not enrolled during all three time periods. Most children were administered the IGDIs three times (mode = 3) and 82.5% of children were assessed two or more times. IGDIs were administered one, two, and three times 17.5%, 25.9%, and 56.6% (respectively) of the children.

Table 1 provides demographic information for participating children. All children met Head Start's criteria for enrollment (Federal Poverty Guidelines and/or special needs designation). The average age of children in this study was 50.5 months in the Fall (age range = 33 – 68 months), 53.6 months in the Winter (range = 31 – 70 months), and 55.7 months in the Spring (range = 33 – 68 months). Ethnic information was not made available to us, although we had information about home language: parents or caregivers indicated that 44% of the children spoke English as their primary language, 27% spoke Spanish, 11% spoke Somali, and 7% spoke Hmong. Overall, participants included 48.8% ELL children and 4.1% of the children in this sample met Head Start's criteria for medical or developmental disability (more than 10% of enrolled children had disabilities, but only 4.3% of these children participated in IGDI assessments).

Table 1

Demographic Information for Participating Children

	Fall	Winter	Spring	**TOTAL**
Number of children	1924	1970	1854	**2306**
Picture Naming	1774	1848	1797	**2279**
Rhyming	1783	1779	1737	**2275**
Alliteration	1782	1747	1703	**2262**
Mean age (months)	50.5	53.6	55.7	**55.0**
Age range (months)	33-68	31-70	33-68	**31-70**
Number of WD* Children (%)	72 (4.1)	79 (4.3)	69 (3.8)	**94 (4.1)**
English speakers	824 (46.4)	846 (45.8)	787 (43.8)	**1006 (44.1)**
Number of ELL** Children (%)	923 (52.0)	988 (53.5)	858 (47.7)	**1113 (48.8)**
Spanish speakers	508 (28.6)	542 (29.3)	534 (29.7)	**613 (26.9)**
Somali speakers	188 (10.6)	209 (11.3)	119 (6.2)	**238 (10.4)**
Hmong speakers	139 (7.8)	145 (7.8)	138 (7.8)	**157 (6.9)**
Other Languages	88 (5.0)	92 (5.0)	67 (3.7)	**105 (4.6)**
Primary language unidentified	27 (1.5)	14 (0.8)	152 (8.5)	**160 (7.0)**
Boys (%)	850 (47.9)	905 (49.0)	801 (44.6)	**1042 (45.7)**
Gender unidentified	16 (0.9)	4 (0.2)	145 (8.0)	**149 (6.5)**

*WD= Children with Disabilities; **ELL=Children Learning English as a 2nd Language

MEASURES

Picture Naming IGDI. Children's expressive language skills were measured with Picture Naming (Missal & McConnell, 2004), an individually administered test developed for use with preschoolers between approximately 30 and 70 months of age. For this task, children are shown cards with a single color picture (photographs and line drawings) of everyday objects with names in the vocabulary of a typical five-year-old child, and asked to name the objects as quickly as possible in one minute. The teacher demonstrates the task for each child by naming four cards quickly, and then asks the child to name the same cards quickly. During administration, if the child is unable to name the sample cards, the task is discontinued. If the child does not respond within 3 seconds, the examiner prompts the child by asking, "What's this?" or "Do you know what this is?" and provides an additional 2 seconds for the child to respond before advancing to the next card. A child's score is the number of pictures named correctly

in one minute. (More information, technical reports, samples, procedures, and stimulus materials are available at http://getgotgo.net).

Psychometric properties of Picture Naming indicate that it is a valid and reliable measure and sensitive to growth for young children. For example, Picture Naming correlates with other standardized measures of language development (McConnell, Priest et al., 2002; Missall, 2002; Priest, Davis, McConnell, McEvoy, & Shin, 1999; Priest, McConnell et al., 2000), is highly correlated with preschoolers' chronological age (McConnell, Priest et al., 2002), and demonstrates adequate one-month alternate forms reliability and test-retest reliability across three weeks (McConnell, Priest et al., 2002).

Rhyming IGDI. Children's phonological awareness skills were measured with Rhyming (Missall & McConnell, 2004), an individually administered test developed for use with preschoolers between 30 and 70 months of age. For this task, children are shown cards on which four color pictures of words familiar to preschoolers are printed, one at the top and three along the bottom. The teacher names each picture and asks the child to find the one that "rhymes or sounds the same as" the top picture. The teacher repeats this procedure for each item and continues for 2 minutes. Before testing formally begins, the examiner teaches the child what "sounds the same" means and demonstrates the task for the child with two sample cards, then gives the child the opportunity to identify the rhyming word for two additional sample cards (Samples 3 and 4). If the child identifies the incorrect rhyme, the teacher provides the correct response. The child is then presented with two more sample cards (Samples 5 and 6) for which the correct response is not provided if the child responds incorrectly. If the child correctly identifies at least two of Samples 5 through 6, the task continues. If the child scores less than two, the task is discontinued and the child's score is recorded as zero. The child's score is the number of correctly identified rhyming words within 2 minutes.

Research on the psychometric properties of this Rhyming task have provided evidence for its use as a valid, reliable, and sensitive measure of growth in preschoolers' phonemic awareness skills. For example, Rhyming correlated moderately with other standardized measures of phonological awareness and early literacy development (McConnell, Priest et al., 2002), test-retest reliability over three weeks was high (Missall & McConnell, 2004b), concurrent validity with Picture Naming and Alliteration (Missall, 2002) and other measures of literacy skills (McConnell, Priest et al., 2002; Missall, 2002). As with Picture Naming, Rhyming has shown evidence of sensitivity to growth of preschoolers' phonological awareness skills (McConnell, Priest et al., 2002; Priest, McConnell et al., 2000).

Alliteration IGDI. Children's phonological awareness skills were measured with Alliteration (Missall & McConnell, 2004), an individually administered test similar to the Rhyming measure developed for use with preschoolers between 30 and 70 months. For this task, the examiner names each of four pictures and asks the child to find one (of three) in the bottom row that "starts with the same sound" as the one top picture. As with Rhyming, before testing formally begins, the teacher demonstrates the task for the child with three sets of sample cards and testing continues only if the child correctly identified at least two of Samples 4 through 6. Again, the child's score is the number of correctly identified words within 2 minutes.

Psychometric characteristics for Alliteration are strong and Alliteration scores appear to be stable over time. For example, Alliteration correlated highly on standardized measures of language development and literacy skills (McConnell, Priest et al., 2002), test-retest reliability was moderate to high (Missall & McConnell, 2004), concurrent validity was demonstrated with other literacy measures (McConnell, Priest et al., 2002; Missall, 2002), and sensitivity to growth was demonstrated with a high correlation between chronological age and the results of the Alliteration IGDIs from typically developing children (McConnell, Priest et al., 2002), children enrolled in Head Start, and children receiving early childhood special education services (Priest, McConnell et al., 2000).

PROCEDURES

Lead researchers and graduate students from the University of Minnesota trained Head Start teachers to administer the three preschool Picture Naming, Rhyming, and Alliteration IGDIs. During a designated two-week period in the fall (late September, early October), teachers individually administered IGDIs to each child in their classroom. Every attempt was made to administer IGDIs to every child enrolled during that period, although some children were absent during the entire period. This procedure was repeated in the winter (late December, early January) and spring (late March, early April) of the 2002-2003 school year. Teachers administered each measure and immediately recorded the child's score on a recording form, which included the child's identification number, gender, date of birth, primary language spoken at home, and disability status. We used this extant data for analysis of emergent language abilities of young children who speak Somali, Hmong, Spanish and English.

RESULTS

ANALYSIS STRATEGY

Analyses examined the extent that three IGDIs measured emergent literacy skills Somali-, Hmong-, Spanish-, and English-speaking preschoolers. The outcome measures were scores on three IGDI measures, Picture Naming, Rhyming, and Alliteration, administered three times over the course of an academic year. We predicted that all children would exhibit growth over time (i.e. that the IGDIs would be sensitive to children's growth in emergent literacy skills). Further, we hypothesized that the PLE children would have higher scores and greater slopes (rates of growth) than the ELL children, but that the ELL children would be similar to each other.

Table 2 provides descriptive statistics for means, standard deviations, and ranges for each of the measures at each assessment point. Mean Picture Naming, Rhyming, and Alliteration scores increased over time for each measure (14.7, 1.7, and 1.0, respectively). To provide descriptive information about the independence among classrooms and Head Start centers, ANOVAs were conducted using the mean scores of the three IGDI measures. ANOVA indicated that classrooms and Head Start centers were significantly different from one another, supporting the need for a statistical procedure that takes into account the lack of independence between repeated observations of each child.

Table 2

Descriptive statistics for Picture Naming, Rhyming, and Alliteration

	Fall	Winter	Spring	**Total**
Mean age	50.5	53.6	55.7	**55.0**
Age range	33-68	31-70	33-68	**31-70**
IGDI	Mean (SD)	Mean (SD)	Mean (SD)	**Mean (SD)**
Picture Naming	12.1 (9.7)	14.9 (10.2)	17.1 (10.4)	**14.7 (10.3)**
range	0-64	0-58	0-50	**0-64**
Rhyming	1.0 (2.5)	1.6 (3.1)	2.4 (4.1)	**1.7 (3.3)**
range	0-19	0-23	0-25	**0-25**
Alliteration	0.6 (1.8)	1.0 (2.2)	1.5 (3.1)	**1.0 (2.3)**
range	0-18	0-20	0-19	**0-20**

Hierarchical Linear Modeling (HLM) is a multilevel modeling procedure that accounts for repeated measures found in studies of children over time and/or for nested data (see Raudenbush & Bryk, 2002, for complete details). HLM was conducted to examine (1) differences in mean scores and rates of growth among English-, Somali-, Hmong-, and Spanish-speaking children's emergent literacy scores and (2) contribution of *Home Language* to explain individual variance among children's emergent literacy scores. We wanted to know if children's IGDI scores (at all three assessments for a particular measure) and change or rate of growth for each IGDI measure (among the three assessments) were related to home language. The repeated measure analyses simultaneously predicted the overall score and rate of growth in children's emergent literacy skills. The repeated measures analyses used the predicted value of IGDI scores for a child at the median age of 53.92 months to be the intercept[1] and the difference between each administration as the slope. The main effect for emergent literacy skills in these analyses tested the extent to which each measure related to the median age of the children, and the interaction of age and IGDI score tested the extent to which that emergent literacy measure related to change over time. These analyses allowed for repeated assessments of both the independent (IGDI measures) and the dependent variables (children's scores). An additional consideration in the analysis of Rhyming and Alliteration scores was that they were positively skewed with modal raw scores at or near zero, considered a Poisson distribution. However, HLM is a robust statistical procedure that can account for the Poisson distribution with a non-linear analysis (Raudenbush & Bryk, 2002). With the HLM application of this non-linear analysis, a unit-specific model was used to represent the expected outcomes for Rhyming and Alliteration with the level-2 variable reduced on a given set of random effects. In addition, a log transformation of expected outcomes was required to account for the Poisson distribution (Raudenbush & Bryk, 2002).

In a follow-up analysis, we wanted to determine whether children's home language contributed over and above the change in scores due to age in predicting their emergent literacy growth. The analysis added Somali, Hmong, Spanish, and English to the model that included age and IGDI scores. When using HLM to model growth, as done here, a null model has limited predictor variables (e.g., age), while a conditional model adds explanatory variables (e.g., language, sex).

[1] In HLM, centering of the intercept is based on theoretical and/or practical concerns. For descriptive purposes, when theoretical and practical considerations do not dictate otherwise, there is some analytic benefit for locating the intercept near the center of available data.

NULL MODEL: EMERGENT LITERACY SKILLS

The Null Model was used to examine the IGDIs' sensitivity to growth and thus used time (age) as a predictor variable. Children's scores based on their age were estimated by modeling the *Age* variable at Level-1 throughout these analyses (see Appendix A for specifications and equations). The first HLM analysis described children's rate of growth for each of the early literacy IGDIs simply based on age. The developmental trajectory consisted of an intercept (estimated score when a child is at the median age of 53.92 months) and a slope (estimated rate of change per month over the three assessment periods). Results, presented in Table 3, indicated that children's Picture Naming, Rhyming, and Alliteration scores increased significantly with age. On Picture Naming, the mean developmental trajectory for all children increased at a rate of 0.61 pictures per month ($t = 27.88$, $p<.001$), and the average score at the median age (53.92 months) was 15.09 ($t = 75.81$, $p<.001$).

Table 3

Growth Curve Analysis: Predicting Trajectories of Preschool IGDI Scores from *Age*

Individual Growth Curves			
	Null Model: Effect of *Age*		
Level-1 Predictor (*Age*)	Variance	df	χ^2
Picture Naming Intercept	63.544	1740	4947.2***
Slope	0.028	1740	1959.2***
Rhyming Intercept	2.362	1740	10872.9***
Slope	0.021	1740	2646.3***
Alliteration Intercept	2.401	1740	8419.2***
Slope	0.017	1740	1942.6***
Group Growth Curves			
	Null Model: Effect of *Age*		
Level-2 Predictors	β	t -ratio	se
Picture Naming Intercept	15.090	75.808***	0.199
Slope	0.607	27.884***	0.020
Rhyming Intercept	- 0.561	12.474***	0.045
Slope	0.149	24.618***	0.006
Alliteration Intercept	- 1.078	-22.389***	0.049
Slope	0.134	20.836***	0.007

*p<.10; **p<.01;***p<.001

On Rhyming, children slowly increased their score at a log rate of 0.15 cards per month (t = 24.62, p<.001). The mean developmental trajectory (i.e. slope) estimated an average score of 0.57 (after the log transformation [i.e. $-.561e^{x}$]; t = 12.47, p<.001) at the median age. Alliteration scores increased at a log rate of 0.13 cards per month (t = 20.84, p<.001). A child's mean developmental trajectory estimated an average score of 0.34 (after the log transformation; t = -22.39, p<.001) at the median age. Thus, children did not start to rhyme or alliterate until approximately 48 months of age, and then their rate of increase was rapid and steady. Thus, this analysis supported the hypothesis that all children would show progress over time on each of the IGDI measures.

Examination of the variance components for Picture Naming revealed a significant variation around the intercept (χ^2 [1740] = 4947.2, p < .001) and slope (χ^2 [1740] = 1959.2, p < .001) yet to be explained. Similar variation occurred for Rhyming and Alliteration (Table 3). These results suggest significant variation among children in intercept and slope on Picture Naming, Rhyming, and Alliteration scores. Thus, modeling the *Age* parameter with additional variables at Level 2 is warranted to understand the developmental trajectories in young children for Naming, Rhyming, and Alliteration.

CONDITIONAL MODEL 1: EMERGENT LITERACY SKILLS AND *HOME LANGUAGE*

The follow-up analysis added the specific home language to the Null Model that included preschool IGDI early literacy scores and age to determine whether home language contributed over and above the change in scores due to age in predicting preschoolers' emergent literacy growth. A separate model was tested for which the Level-2 equation was expanded from the Null Model to:

$$\pi_0 = \beta_{00} + \beta_{01}(Somali) + \beta_{02}(Spanish) + \beta_{03}(Hmong) + \beta_{01}(Other\ Language) + r_0$$
$$\pi_1 = \beta_{10} + \beta_{11}(Somali) + \beta_{12}(Spanish) + \beta_{13}(Hmong) + \beta_{14}(Other\ Language) + r_1$$

For this analysis, the intercept (β_{00}) represented the mean IGDI score for children whose primary language was English (PLE) at the median age of 53.92 months, and β_1 through β_4 represented children whose primary language was Somali, Spanish, Hmong, or Other Languages, respectively. Results are presented in Tables 4, 5, and 6.

After adjusting for age for Picture Naming, the intercept for PLE children was 15.14 words per minute and PLE children scored significantly higher than children who speak *Somali* (β_1 = -6.49, t = -11.69, p < .001), *Spanish*, (β_2 = -10.51, t = 26.94, p < .001), *Hmong*, (β_3 = -10.04, t = -15.73, p < .001), and *Other Languages* (β_4 = -4.41, t = -4.25, p < .001). Rate of growth for PLE children was .61 pictures per month (t = 30.72, p < .001), and rate of growth for the other language groups was not significantly different from PLE children. Growth rates varied from -.05 pictures per month for *Other Languages* to .08 pictures per month for *Somali* children (*Somali* t = 1.28, p = 0.20; *Spanish* t = -0.77, p = 0.44; *Hmong* t = -0.23, p = 0.82; *Other Language* t = -0.43, p = 0.66; Table 4). Thus, as Figure 1a illustrates, ELL children were older than PLE children when they began naming pictures in English, but rates of growth for all groups were similar. Thus, analysis of Picture Naming supports the hypothesis that PLE children would have higher scores than ELL children. However, this analysis did not support the hypothesis that PLE children would have greater rates of growth than ELL children since all children had similar growth trajectories.

Table 4

Growth Curve Analysis: Predicting Trajectories of Preschool Picture Naming IGDI Scores from *Language*

Individual Growth Curves				
	Model 1: Effect of Language			
Level-1 Predictor (*Age*)	Variance	df	χ^2	PV[a]
Picture Naming Intercept	40.200	1736	3720.6***	0.367
Slope	0.049	1736	1974.5***	-0.751
Group Growth Curves				
	Model 1: Effect of Language			
Level-2 Predictors	β	t -ratio	se	
Picture Naming Intercept (English)	15.139	89.889***	0.168	
Somali	-6.486	-11.685***	0.555	
Spanish	-10.505	-26.936***	0.390	
Hmong	-10.042	-15.734***	0.638	
Other Language	-4.408	-4.250***	1.037	
Picture Naming Slope (English)	0.612	30.718***	0.020	
Somali	0.083	1.278	0.065	
Spanish	0.036	-0.769	0.047	

Hmong	-0.017	-0.228	0.076
Other Language	-0.049	-0.434	0.112

[a] Proportion of variance explained by Level-2 predictors.

*p<.10; **p<.01;***p<.001

The variance components of random effects showed significant individual differences yet to be explained in both intercept and slope (Table 4). Comparing variance estimates between the Null Model and Conditional Model 1, the proportion of variance in the intercept and slope explained by home language was 0.37 and -0.75, respectively. These proportions indicated a child's primary language explained much of the variability in children's Picture Naming scores, although home language did not contribute significantly more than age to variability in rate of growth. Thus, *Home Language* and *Age* interacted in rate of growth.

Table 5

Growth Curve Analysis: Predicting Trajectories of Preschool Rhyming IGDI Scores from *Language*

Individual Growth Curves				
	Model 1: Effect of Language			
Level-1 Predictor (*Age*)	Variance	df	χ^2	PV[a]
Rhyming Intercept	1.999	1736	8839.3***	0.154
Slope	0.019	1736	2632.4***	0.085
Group Growth Curves				
	Model 1: Effect of Language			
Level-2 Predictors	β	t -ratio	se	
Rhyming Intercept (English)	-0.626	-14.067***	0.045	
Somali	-1.048	-6.962***	0.151	
Spanish	-1.626	-15.116***	0.108	
Hmong	-2.042	-9.578***	0.213	
Other Language	-0.280	-1.417	0.198	
Rhyming Slope (English)	0.158	25.428***	0.006	
Somali	0.048	2.189 **	0.022	
Spanish	0.021	1.389	0.015	
Hmong	0.073	2.434 *	0.030	

Other Language	0.016	0.567	0.028

[a] Proportion of variance explained by Level-2 predictors.

*p<.10; **p<.01;***p<.001

Mean Rhyming scores for the PLE children were significantly higher than *Somali* (β_1 = -1.048, log transformation = 0.187, t = -6.96, p < .001), *Spanish* (β_2 = -1.63, log transformation = 0.105, t = -15.12, p < .001), and *Hmong* (β_3 = -2.04, log transformation = 0.069, t = -9.58, p < .001) groups, but not significantly different from children in the *Other Language* group (β_4 = -0.280, log transformation = 0.404, t = -1.42, p = 0.16; Table 5). Rate of growth in Rhyming for PLE children (β_0 = 0.16, log transformation = 1.171) was less than for ELL children, although the difference was significant only for the *Somali-* (β_1 = 0.05, log transformation = 1.049, t = 2.19, p < .10) and *Hmong*-speaking children (β_3 = 0.07, log transformation = 1.075, t = 2.43, p < .10; Table 5). Upward trajectories of Rhyming skills began earliest for *English*-speaking children, and started latest for *Hmong*-speaking children. Children who speak *Other Languages* displayed a trajectory very similar to PLE children (Figure 1b).

FIGURE LEGEND

Figure 1a-1c: Rate of growth for IGDIs based on *Language*. Illustrates differences in average Picture Naming, Rhyming, and Alliteration scores and rates of growth for English-, Somali-, Spanish-, and Hmong-speaking children in this sample.

Figure 1: Rate of growth for Picture Naming, Rhyming, and Alliteration based on *Age*

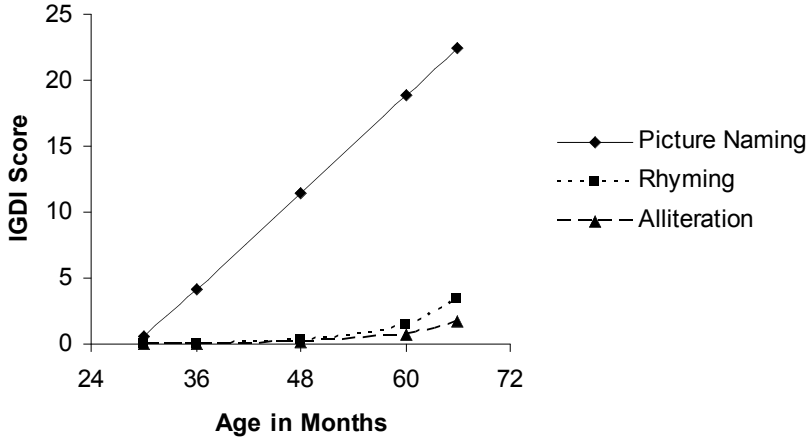
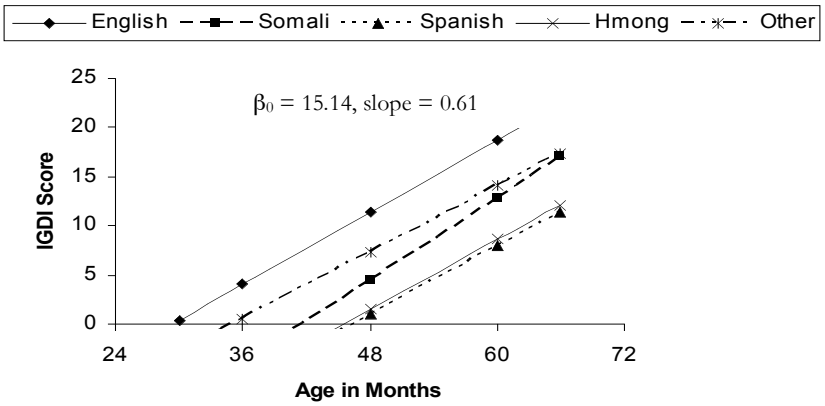
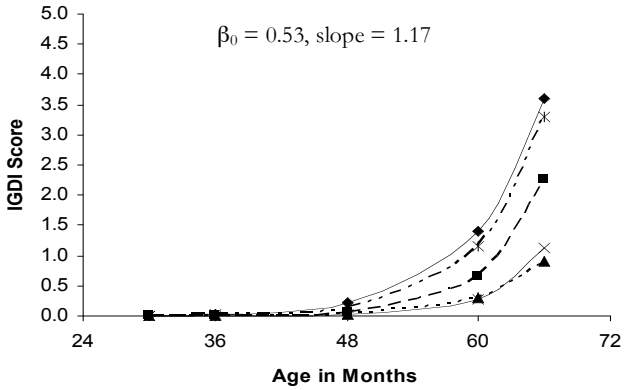


Figure 2: Rate of growth for IGDIs based on *Language*

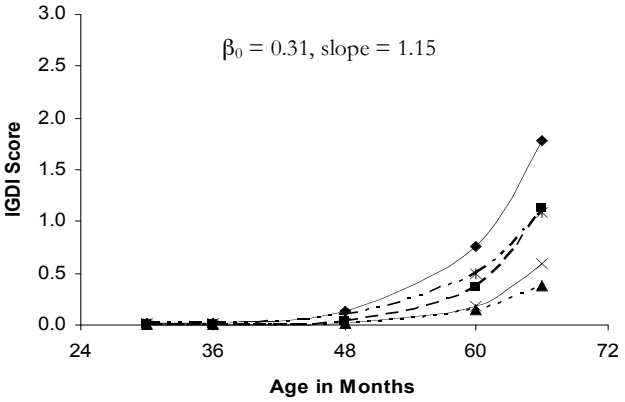
2a) Picture Naming



2b) Rhyming



2c) Alliteration



As Table 6 shows, mean Alliteration scores for PLE children at the median age were significantly higher than all ELL groups except for the *Other Language* group (*Somali* = -0.99, log transformation = 0.117, t = -5.93, p < .001; *Spanish* = -1.65, log transformation = 0.060, t = -13.59, p < .001; *Hmong* = -1.74, log transformation = 0.055, t = -7.62, p < .001; *Other Language* = -0.32, log transformation = 0.228, t = -1.50, p = 0.13). Rate of growth of PLE children was less than *Somali*, *Spanish*, and *Hmong* children's rate of growth, although the difference was significant only for *Somali*- and *Hmong*-speaking children (*Somali* = 0.04, log transformation = 1.045, t = 1.84, p < .10; *Spanish* = 0.01, log transformation = 1.010, t = 0.60, p = 0.55; *Hmong* = 0.05, log transformation = 1.056, t = 1.75, p < .10; Table 6). Rate of growth for the *Other Language* group (*Other Language* = -0.01, log transformation = 0.986, t = -0.50, p = 0.61) was smaller than for PLE children. PLE and the *Other Language* group started exhibiting Alliteration skills before the other ELL groups. However, rate of growth of the *Other Language* group slowed down as *Somali* children's rate increased. *Spanish*-speaking children started displaying Alliteration skills later than most groups and their rate of growth was slower and with a slope similar to *English*-speaking children, who began Alliteration earlier (Figure 1c). Thus, analyses of Rhyming and Alliteration support the hypothesis that PLE children would have higher scores than ELL children, and that mean scores of ELL would be similar. These analyses did not support the hypothesis that PLE children would have greater rates of growth than ELL children, or that the performance of all ELL children would be similar. Rather, ELL children exhibited heterogeneous rates of growth, with Somali and Hmong children acquiring rhyming and alliteration skills at a faster rate than English- or Spanish-speaking children.

Table 6

Growth Curve Analysis: Predicting Trajectories of Preschool Picture Naming IGDI Scores from *Language*

Individual Growth Curves				
	Model 1: Effect of Language			
Level-1 Predictor (*Age*)	Variance	df	χ^2	PV[a]
Alliteration Intercept	2.105	1736	6906.6***	0.125
Slope	0.015	1736	1957.0***	0.085
Group Growth Curves				
	Model 1: Effect of Language			
Level-2 Predictors	β	t -ratio	se	
Alliteration Intercept (English)	-1.161	-23.575***	0.049	
Somali	-0.988	-5.929***	0.167	
Spanish	-1.651	-13.591***	0.122	
Hmong	-1.744	-7.618***	0.229	
Other Language	-0.318	-1.499	0.212	
Alliteration Slope (English)	0.144	21.266***	0.007	
Somali	0.044	1.836 *	0.024	
Spanish	0.010	0.601	0.017	
Level-2 Predictors	β	t -ratio	se	
Hmong	0.054	1.745 *	0.031	
Other Language	-0.014	-0.503	0.029	

[a] Proportion of variance explained by Level-2 predictors.

*p<.10; **p<.01;***p<.001

Variance estimates from the Null Model were compared with those in Conditional Model 1 for Rhyming and Alliteration. For Rhyming, proportion of variance in intercept and slope explained by home Language was 0.15 and .08, respectively (Table 5). For Alliteration, proportion of variance in intercept and slope explained by home Language was .125 and .085, respectively (Table 6). These proportions indicated that a fair amount of variability among individual children's mean Rhyming and Alliteration

scores and rate of growth was explained by their home language.[2]

DISCUSSION

This study followed English-, Somali-, Hmong-, and Spanish-speaking preschoolers from low-income families for nine months, and examined their performance on three IGDI emergent literacy measures, the sensitivity of the IGDIs to measure growth of emergent literacy skills, and relations between the different languages on children's performance. All language groups showed significant progress in mean scores and rates of growth across all three measures. When children's performance was examined over and above their ages, we found that differences in mean scores for Picture Naming were due primarily to their home language, but rates of growth were related to their age. However, differences in both mean scores and rates of growth for Rhyming and Alliteration varied due to home language. Not surprisingly, English-speaking children had higher mean scores for Rhyming and Alliteration across all time periods than children whose primary language was not English. Yet, although rates of growth were significant for all children on all measures, Somali and Hmong children demonstrated significantly greater rates of growth on Rhyming and Alliteration than English-speaking children. Thus, we found that home *Language* is acting as a mediator in early literacy development (Baron & Kenny, 1986). The rule of parsimony would dictate that home *Language* is a primary variable to consider and include in a model that explains variability in English early literacy skills among children in poverty.

Aside from examining mean scores and rates of growth for children with diverse home languages, an additional objective of this research was to examine the sensitivity of the Picture Naming, Rhyming, and Alliteration IGDIs in measuring growth for young children whose primary language was not English. Examination of Picture Naming scores indicates that it is a sensitive measure of growth for children who speak Somali, Hmong, and Spanish. Results of this study, plus past research that has demonstrated sensitivity to growth for children in typical early childhood classrooms and English-speaking children in Head Start, lends compelling support for the use of the Picture Naming IGDIs as a

[2] In order to understand the contribution and examine possible mediating effects of these child variables with sex of children on emergent literacy skills, we modeled Sex (Conditional Model 2) and *Sex* and *Language* (Full Model). Conditional Model 2 and the Full Model explained nothing more than Conditional Model 1 with *Language* as the only variable.

way to monitor the oral language and/or emergent literacy skills of ELL children. Rhyming and Alliteration, however, do not provide such strong evidence of sensitivity. It appears that Rhyming and Alliteration scores for PLE and ELL children display very little growth between 30 and 48 months of age. This may be an indicator that the properties of Rhyming and Alliteration are not sensitive enough to capture growth of phonemic awareness skills over a three- to six-month period of time for ELL children. Thus, Rhyming and Alliteration in English may not be appropriate measures of emergent literacy skills for children who speak a different home language, or these measures might be more appropriate for children older than 48 months. Future research should explore the possibility of modifying Rhyming and Alliteration so they are more sensitive measures of emergent literacy skills for children whose first language is not English, or develop alternatives measures of early emergent literacy skills.

Differences in average scores between PLE and ELL children are not surprising given available research (Adams, 1990; Durgunoglu, Mir, & Arino-Marti, 2002; Snow, C., 1999; Tabors, 1998; Tabors & Snow, 2002) and clinical experience. However, this study extends past research by describing rates of growth of the two groups, as well as differences in average scores and rates of growth among Somali-, Hmong-, and Spanish-speaking preschoolers. But how are these differences between Somali-, Spanish-, and Hmong-speaking children explained, and what are the implications for optimally serving these children and preparing them for successful reading experiences?

There are several possible explanations for why home language may result in differences in early literacy trajectories. First, many researchers report that differences between language groups is influenced by family characteristics such as predominant language spoken in the home, whether literacy activities in the home and preschool occur in the child's primary language or English, the length of time children have been exposed to their primary language and English, how well parents speak English, and parental educational expectations (Goldenberg, Gallimore, Reese, & Garnier, 2001; Hart & Risley, 1995; Whitehurst & Fischel, 2000). One study compared home literacy experiences of Spanish-speaking children who either learned English simultaneously with Spanish, or only when they entered preschool (sequentially). Results revealed no differences in the home literacy experiences of the two groups related to the value placed on literacy, although mothers of children who were learning English and Spanish simultaneously engaged in more activities that supported academic achievement than mothers of sequential learners (Hammer, Miccio, & Wagstaff, 2003). Related to family characteristics, research findings by Roberts, Jurgens, and Burchinal (2005) indicated that the overall quality and responsiveness of the home environment of African-American children was the most consistent and strongest predictor of children's literacy skills. Thus, we have two studies that suggest that family characteristics and quality

of the home environment vary with the cultural/linguistic groups. The present study extends this research and provides evidence that home environment and linguistic group interact in ways that affect children's emergent literacy skills. Future research should include variables for family characteristics and the home environment for children with diverse linguistic backgrounds, including the bilingual/multilingual language experiences that each child has had, to examine their effect on the development of early literacy skills for this group of children.

A second explanation for the differences in language groups may be related to the match between early literacy activities at home and those in school. Several researchers posit that this home-school match influences children's early literacy and later reading success (Gee, 2002; Pellegrini, 2002). It may be that the early literacy home environment of Somali children in this sample, who developed phonemic awareness skills at a faster rate than their Spanish or Hmong counterparts, is more similar to the early literacy environment at Head Start. Pursuing this theory could have implications for adapting parenting education opportunities and for adapting some classroom environments in center-based early childhood education settings. There may also be merit in longitudinal ethnological research that considers both pre-immigration cultural variables and post-immigration acculturation as process variables influencing children's development. A third possibility for the differences in emergent literacy skills being mediated by home language may be related to teachers' primary language and their match to children's language, and how and when the two languages are used during the preschool day (Limbos & Geva, 2001; Tabors & Snow, 2002). In the program where these data were collected, if more than half of the children in a classroom spoke the same primary language other than English, the primary language of a member of the classroom team (often an assistant teacher) was matched to the children's primary language. Nonetheless, some ELL children were not matched with an educator who spoke the same primary language. In either scenario, teacher language differences might explain group language differences. Future research should be conducted to assess whether systematic teacher-child language matching affects the rate of growth in children's early literacy skills. This research could offer compelling evidence about the importance of building first language skills before attempting to construct early literacy awareness in a second language.

A fourth possible explanation for differences among language groups may be related to cross-linguistic transfer (literacy skills learned in one language affect acquisition of skills in another language). Research has demonstrated similar predictors of literacy development (phonological awareness, letter knowledge) in other monolingual populations for Spanish (Durgunoglu, Nagy, & Hansin-Bhatt, 1993), French (Sprenger-Charolles, Siegel, & Bonnet, 1998), Portuguese (Cardoso-Martins, 1995), Turkish

(Durgunoglu & Oeney, 2000; Oeney & Durgunoglu, 1997), Italian (Cossu, Shankweiler, Liberman, Katz, & Tola, 1988), and German (Wimmer, Landerl, Linortner, & Hummer, 1991). Additionally, numerous studies have found correlations across Spanish, Chinese, and Hmong languages in phonological awareness (Durgunoglu, 1998; Gottardo, 2002; Gottardo, Yan, Siegel, & Wade-Wooley, 2001; Lindsey, Manis, & Bailey, 2003). Although no published research to date has reported on this phenomenon for speakers of Somali, it is possible that they also experience cross-linguistic transfer. Future research to explore cross-linguistic transfer in other languages is warranted, as well as research to compare languages to see if degrees of transfer vary with the primary language. This information could offer guidance about differentiated intervention strategies based on language characteristics.

For several reasons, it is important to interpret these data cautiously. First, the results should be considered carefully when generalizing to other samples. Although the sample size was large, it included only children from poverty. Furthermore, we lacked information about the English or home language proficiency of the ELL children. Different levels of proficiency in either language may result in different trajectories for each language group (Tabors & Snow, 2002). Accounting for relations among Picture Naming, Rhyming, and Alliteration IGDIs might be further enhanced if children were assessed in English and their home language. Future research should examine other variables that may have equal or greater affects on early literacy trajectories of children in poverty (e.g., proficiency, language interactions, home environment, interventions) and compare them to the trajectories of children not in poverty. Such research could provide information for policy makers and interventionists about the neediest groups to target and where resources should be directed.

Second, although the large sample size reduces the variance due to error, variance due to teacher bias during test administration is still possible. Because we used extant data, we did not have the opportunity to control for fidelity of administration of the IGDIs or to evaluate reliability of administration. The psychometric properties of the IGDIs provide evidence for their reliability with trained and closely supervised examiners. Test-retest reliability with a small sample of Head Start teachers was also good. However, we do not know how the examiner's sophistication (e.g., education and experience) or level of training for IGDI administration would affect the reliability of administration. As IGDIs become more widely-used, how the reliability of results is affected by the fidelity of administration would be an important matter to address.

In closing, this study found that the Picture Naming IGDI is a sensitive and useful measure to monitor the oral English-language development and emergent literacy skills of ELL children, and that

Rhyming and Alliteration IGDIs are not as sensitive. In addition, this study found that home language is a powerful correlate of emergent literacy skills among Head Start preschool children. Significant differences in mean scores of all ELL language groups indicate that all Head Start preschoolers are acquiring English, some at a greater rate than others. While these data suggest that not all ELL children are closing the gap with PLE children, further research about the differences in performance between ELL and PLE children in other socioeconomic groups, or about other children in poverty who do not participate in Head Start is surely warranted. Furthermore, closing that gap between ELL and PLE performance on the IGDIs is clearly an outcome that early childhood educators should pursue. Policy makers, educational administrators, and educators who are concerned with closing the gap between children in poverty, which includes many children learning English, must recognize the challenges of serving our poorest children. Future research to identify other prominent variables that impact early literacy development and modify existing tools, as well as a continuing commitment to resources that will support programs for children in poverty and children learning English, can only promote the advancement of early literacy skills for all children.

REFERENCES

Adams, M. J. (1990). *Learning to read: Thinking and learning about print.* Cambridge, MA: MIT Press.

American Speech-Language-Hearing Association. (2001). *Roles and responsibilities of speech-language pathologists with respect to reading and writing in children and adolescents* [Technical Report]. Retrieved on May 15, 2007 from www.asha.org/policy.

Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of personality and social psychology, 51*(6), 1173-1182.

Beaulieu, L. M. (2002). African American children and literacy: Literacy development in the early childhood years. In S. J. Denbo & L. M. Beaulieu (Eds.), *Improving schools for African American students: A reader for educational leaders* (pp. 125-131). Springfield, IL: Charles C. Thomas Publisher, Ltd.

Bricker, D., Pretti-Frontczak, K., & McComas, N. (2002). *An activity-based approach to early intervention.* Baltimore: Paul H. Brookes Publishing Co.

Cardoso-Martins, C. (1995). Sensitivity to rhymes, syllables, and phonemes in literacy acquisition in portuguese. *Reading research quarterly, 30*(4), 808-828.

ChildStats.gov. (2005). Population and family characteristics. Retrieved on May 15, 2007 from www.childstats.gov/amchildren05/pop5.asp.

Cossu, G., Shankweiler, D., Liberman, I. S., Katz, L., & Tola, G. (1988). Awareness of phonological segments and reading ability in Italian children. *Applied psycholinguistics, p*, 1-16.

Deno, S. L. (1986). Formative evaluation of individual student programs: A new role for school psychologists. *School psychology review, 15*, 358-374.

Deno, S. L. (1997). Whether thou goest...Perspectives on progress monitoring. In J. W. Lloyd, E. J. Kameenui & D. Chard (Eds.), *Issues in educating students with disabilities* (pp. 77-99). Mahwah, NJ: Lawrence Erlbaum Associates.

Deno, S. L., Mirkin, P. K., & Chiang, B. (1982). Identifying valid measures of reading. *Exceptional children, 49*(1), 36-45.

Durgunoglu, A. Y. (1998). Acquiring literacy in English and Spanish in the united states. In A. Y. Durgunoglu & L. Verhoeven (Eds.), *Literacy development in a multicultural context: Cross-cultural perspectives* (pp. 135-146). Mahwah, NJ: Erlbaum and Associates.

Durgunoglu, A. Y., Mir, M., & Arino-Marti. (2002). The relationship between bilingual children's reading and writing in their two languages. In S. Ransdell & M. L. Barbier (Eds.), *Psycholinguistic approaches to understanding second-language writing* (pp. 81-100). Dordrecht, The Netherlands: Kluwer.

Durgunoglu, A. Y., Nagy, W. E., & Hansin-Bhatt, B. J. (1993). Cross-language transfer of phonological awareness. *Journal of educational psychology, 85*, 453-465.

Durgunoglu, A. Y., & Oeney, B. (2000). *Literacy development in two languages: Cognitive and sociocultural dimensions of cross-language transfer*. Paper presented at the US Department of Educations, Office of Bilingual Education and Minority Language Affairs, Reading Research Symposium, Washington, DC.

Early Childhood Research Institute on Measuring Growth and Development (ECRI). (1998, 2000). *Research and Development of Exploring Solutions Assessments for Children Between Birth and Age Eight* (Technical Report 5)., University of Minnesota, Minneapolis. Retrieved June 25, 2003 from http://ggg.umn.edu/pdf/ecrirpt5.pdf

Fey, M., Catts, H., & Larrivee, l. (1995).Preparing preschoolers for the academic and social challenges of school. In M. Fey, J. Windsor, & S. Warren (Eds.), *Language intervention: Preschool through the elementary years* (pp. 3-37). Baltimore, MD: Brookes.

Fuchs, L. S., & Deno, S. L. (1991). Paradigmatic distinctions between instructionally relevant measurement models. *Exceptional children, 57*, 488-500.

Gee, J. P. (2002). A sociocultural perspective on early literacy development. In S. B. Neuman & D. K. Dickenson (Eds.), *Handbook of early literacy research*. New York: Guilfold Press.

Goldenberg, C., Gallimore, R., Reese, L., & Garnier, H. (2001). Cause or effect: A longitudinal study of immigrant Latino parents' aspirations and expectations, and their children's school performance. *American Educational Research Journal, 38*, 547-582.

Good, R. H., Simmons, D. C., & Smith, S. B. (1998). Effective academic interventions in the United States: Evaluating and enhancing the acquisition of early reading skills. *School psychology review, 27*(1).

Gottardo, A. (2002). The relationship between language and reading skills in bilingual Spanish-speakers. *Topics in language disorders, 22*, 46-70.

Gottardo, A., Yan, B., Siegel, L. S., & Wade-Wooley, L. (2001). Factors related to English reading performance in children with Chinese as a first language: Evidence of cross-linguistic transfer of phonological processing. *Journal of educational psychology, 93*, 530-542.

Hammer, C., Miccio, A., & Wagstaff, D. (2003). Home literacy experiences and their relationship to bilingual preschoolers' developing English literacy abilities: An initial investigation. *Language, speech, and hearing services in the schools, 34*, 20-30.

Hart, B., & Risley, T. (1995). *Meaningful differences in the everyday experience of young American children.* Baltimore: Paul H. Brookes Publishing Co.

Limbos, M., & Geva, E. (2001). Accuracy of teacher assessments of ESL children at-risk for reading disability. *Journal of learning disabilities, 34*, 136-151.

Lindsey, K. A., Manis, F. R., & Bailey, C. E. (2003). Prediction of first-grade reading in Spanish-speaking English-language learners. *Journal of educational psychology, 95*, 482-494.

Lombardino, L., Kieberman, R.J., & Brown, J.C. *Assessment of Literacy and Language.* San Antonio, TX: Harcourt Assessment.

Lonigan, C.J., Wagner, R.K., & Torgeson, J.K. (2007). *Test of Preschool Early Literacy.* Greenville, SC: Super Duper Publications.

McConnell, S.R., McEvoy, M.A., & Priest, J.S. (2002). "Growing" measures for monitoring progress in early childhood education: A research and development process for individual growth and development indicators. *Assessment for effective intervention, 27*(4), 3-14.

McConnell, S. R., Phaneuf, R., & Murphy, L. (2002). *Extending language and literacy assessment to diverse populations: IGDI training and evaluation.* Paper presented at the Council for Exceptional Children, Division of Early Childhood, San Diego, CA.

McConnell, S. R., Priest, J. S., Davis, S. D., & McEvoy, M. A. (2002). Best practices in measuring growth and development in preschool children. In A. Thomas & J. Grimes (Eds.), *Best practices in school psychology* (Vol. 4th edition, pp. 1231-1246). Washington, DC: National Association of School Psychologists.

Missall, K. N. (2002). *Reconceptualizing school adjustment: A search for intervening variables.* US: University Microfilms International.

Missall, K. N., & McConnell, S. R. (2004). *Psychometric characteristics of individual growth and development indicators: Picture naming, rhyming, and alliteration* (Technical Report 8). Retrieved June 20, 2005 from University of Minnesota, Minneapolis, Get It, Got It, Go Website: http://ggg.umn.edu/pdf/ecrirpt8.pdf

Missall**,** K.N., McConnell, S.R., Cadigan, K. *(*2006). Early literacy development: Skill growth and relations between classroom variables for preschool children**.** *Journal of early intervention, 29(1), 1-21.* Council for Exceptional Children.

Nancollis, A., Lawrie, B.A., & Dodd, B. (2005). Phonological awareness intervention and the acquisition of literacy skills in children from deprived social backgrounds. *Language, speech, and hearing services in schools, 36*, 325-335.

Neisworth, J. T., & Bagnato, S. J. (1996). Assessment for early intervention: Emerging themes and practices. In S. L. Odom & M. E. McLean (Eds.), *Early intervention/early childhood special education: Recommended practices* (pp. 23-57). Austin, TX: PRO-ED.

Nicholson, T. (1997). Closing the gap on reading failure: Social background, phonemic awareness, and learning to read. In B. Blachman, A (Ed.), *Foundations of reading acquisition and dyslexia: Implications for early intervention* (pp. 381-407). Mahwah, NY: Lawrence Erlbaum Associates, Publishers.

Nitsiou, C. T. (2001). *The use of the picture naming individual growth and development indicator with young English language learners.* US: University Microfilms International.

Oeney, B., & Durgunoglu, A. Y. (1997). Beginning to read in Turkish: A phonologically transparent orthography. *Applied psycholinguistics, 18*(1), 1-15.

Pellegrini, A. D. (2002). Some theoretical and methodological considerations in studying literacy in social context. In S. B. Neuman & D. K. Dickenson (Eds.), *Handbook of early literacy research* (pp. 54-65). New York: The Guilford Press.

Priest, J. S., Davis, K. N., McConnell, S. R., McEvoy, M. A., & Shin, J. (1999). *Individual growth and development indicators for preschoolers' "expressing meaning" skills: Follow that trajectory!* Paper presented at the Annual conference on the Division for Early Childhood, Council for Exceptional Children, Washington, DC.

Priest, J. S., McConnell, S. R., McEvoy, M. A., & Shin, J. (2000). *Early childhood research institute on measuring growth and development: Progress in five domains.* Paper presented at the Paper presented at the annual conference of the Division for Early Childhood, Council for Exceptional Children, Albuquerque, NM.

Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd Edition ed.). Thousand Oaks, CA: Sage Publications.

Reid, D.K., Hresko, W.P., & Hammill, D.D. (2001). *Test of Early Reading Ability-Third Edition.* Austin, TX: ProEd Publications.

Roberts, J., Jurgens, J., & Burchinal, M. (2005). The role of home literacy practices in preschool children's language and emergent literacy skills. *Journal of speech, language, and hearing research, 48*, 245-359.

Robertson, C., & Slater, W. (1997). *The Phonological Awareness Test.* Chicago: LinguiSystems.

Snow, C. E. (1999). Facilitating language development promotes literacy learning. In L. Eldering & P. P. M. Leseman (Eds.), *Effective early education: Cross cultural perspectives.* New York: Falmer Press.

Snow, C. E., Burns, M. S., & Griffin, P. (1998). *Preventing reading difficulties in young children.* Washington, DC: National Academy Press.

Sprenger-Charolles, L., Siegel, L. S., & Bonnet, P. (1998). Reading and spelling acquisition in french: The role of phonological mediation and orthographic factors. *Journal of experimental child psychology, 68*(2), 134-165.

Tabors, P. O. (1997). Using communication and classroom organization to support second language learners. In P. O. Tabors (Ed.), *One child, two languages: A guide for preschool educators of children learning English as a second language*. Baltimore: Paul H. Brookes Publishing Company.

Tabors, P. O. (1998). What early childhood educators need to know: Developing effective programs for linguistically and culturally diverse children and families. *Young Children, 53*(6), 20-26.

Tabors, P. O., & Snow, C. (2002). Young bilingual children and early literacy development. In S. B. Neuman & D. K. Dickenson (Eds.), *Handbook of early literacy research* (pp. 159-178). New York: The Guilford Press.

Good start, grow smart: The Bush administration's early childhood initiative (2002). Retrieved 11/19/2003 from The White House Website: http://www.whitehouse.gov/infocus/earlychildhood/earlychildhood.html

Whitehurst, G. J. (2000). Reading and language impairments in conditions of poverty. In D. V. M. Bishop & L. B. Leonard (Eds.), *Speech and language impairments in children: Causes, characteristics, intervention and outcome* (pp. 53-71). East Sussex: Psychology Press.

Whitehurst, G. J., & Fischel, J. E. (2000). Reading and language impairments in conditions of poverty. In D. V. M. Bishop & L. B. Leonard (Eds.), *Speech and language impairments in children: Causes, characteristics, intervention and outcome*. Philadelphia, PA: Psychology Press.

Whitehurst, G. J., & Lonigan, C. j. (1998). Child development and emergent literacy. *Child development, 69*(3), 848-872.

Whitehurst, G. J., & Lonigan, C. j. (2002). Emergent literacy: Development from prereaders to readers. In S. B. Neuman & D. K. Dickenson (Eds.), *Handbook of early literacy research* (pp. 11-29). New York: The Guilford Press.

Wimmer, H., Landerl, K., Linortner, R., & Hummer, P. (1991). The relationship of phonemic awareness to reading acquisition: More consequence than precondition but still important. *Cognition, 40*(3), 219-249.

APPENDIX

NULL MODEL EQUATIONS

Under specifications for the Null Model, the coefficient for the intercept (β_0) was the expected IGDI score for child j, and β_1 was the coefficient representing the change in score for child j's age in months. Child *Language* was modeled at Level-2 as a between-children variable. *Language* included English, Spanish, Somali, Hmong, and Other Languages (included Oromo, Vietnamese, Laotian, Swahili). Level-2 variables reflect differences between children and are treated as unchanging with age. Thus, equations for the Null Level-2 model are represented by

$$\pi_0 = \beta_{00} + r_0$$

$$\pi_1 = \beta_{10} + r_1,$$

where β_{00} represented the expected IGDI score for a child, and β_{10} represented the change in the score for a particular child's age in months.